A Formal Negative Result for the Passage from Step (xi-e) to Step (xi-f) in IUT III

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Abstract

We isolate a precise formal claim behind the passage from Step (xi-e) to Step (xi-f) in the proof of Corollary 3.12 of Mochizuki's *Inter-universal Teichmüller Theory III*. The claim is the implication from a purely qualitative package—input prime-strip linkage, simultaneous holomorphic expressibility, algorithmic parallel transport, passage to a common container, and the description of the output log-volumes as a closed half-line—to the numerical bridge statement

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}.$$

We prove a negative result: after formalizing that qualitative package without building in any numerical bridge axiom, the bridge statement is *not* a logical consequence. The proof is by an explicit countermodel in which the input and output objects have the same abstract label and live in the same common container, while the concrete real-valued log-volume of the input lies strictly above the upper end of the output half-line. We also prove that the "approximation" escape hatch mentioned in Step (xi-f) does not weaken the required bridge in the closed-half-line setting: for a closed half-line, approximate realization from below is equivalent to exact membership. This note does *not* disprove IUT. It proves only that the bridge needed for Corollary 3.12 does not follow from the extracted qualitative package alone.

1 Introduction

The end of Step (xi) in the proof of Corollary 3.12 of IUT III is one of the central points of dispute in the literature on inter-universal Teichmüller theory. Mochizuki explicitly says there that, for the purpose of Step (xi), he is retaining only the "qualitative logical aspects/consequences" of Theorem 3.11, namely the properties labeled input prime-strip link (IPL), simultaneous holomorphic expressibility (SHE), and algorithmic parallel transport (APT) [1, pp. 160–162, 180–183]. At the same time, the same paper also warns that the "possible regions" produced by the multiradial algorithm are not automatically comparable with codomain-side structures, and that further techniques are required for such comparisons [1, p. 161; pp. 133–135]. In the proof of Corollary 3.12, Mochizuki nevertheless passes from the statement that the output possible log-volumes are linked/related to the input log-volume to the inclusion

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|},$$

and hence to the inequality

$$-|\log(q)| \leq -|\log(\Theta)|$$

which he says then follows formally [1, p. 183].

In later expository work, Mochizuki reformulates the same underlying picture in terms of "noninterference" and "cohabitation/coexistence" of the q-pilot and Θ-pilot inside a common container obtained by multiradial representation, holomorphic hull, and log-volume [2, p. 98]. Scholze and Stix, on the other hand, single out this same passage as the critical step and object that the required move from abstract linkage to concrete numerical comparison has not been made explicit enough [3, pp. 8–9].

The present note does not attempt to settle the full mathematical dispute over IUT. Instead, it isolates a very specific and formally clean statement:

Question. Suppose one retains only the qualitative package highlighted in the relevant passages—abstract linkage, simultaneous executability, common-container cohabitation, and the fact that the output possible log-volumes form a closed half-line. Does the bridge

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}$$

already follow?

Our answer is negative. We define a formal theory $\mathcal{Q}$ that captures exactly this qualitative package, without adding any new numerical bridge axiom. Then we construct an explicit model of $\mathcal{Q}$ in which the bridge fails. As a consequence,

$$\mathcal{Q} \nvdash \mathbf{Br},$$

where **Br** denotes the desired bridge statement. The proof is elementary but conceptually important: it shows that the bridge is not obtained merely from abstract identification, cohabitation, or simultaneous executability. To derive the inequality, one needs an additional lemma that actually connects the input real value to the output half-line.

The note is organized as follows. In Section 2, we record the precise qualitative package extracted from the source passages. In Section 3, we formalize that package. In Section 4, we prove that in the closed-half-line setting the "approximation" language of Step (xi-f) is equivalent to exact membership. In Section 5, we construct the countermodel and prove the main negative theorem. Finally, in Section 6, we explain what a repaired proof would have to add.

2 Source extraction and scope

For the convenience of the reader, we summarize the source passages that motivate our formalization.

(S1) In Remark 3.11.1 and the beginning of Step (xi), Mochizuki says that, for the application to Corollary 3.12, one may focus only on the qualitative consequences of Theorem 3.11, namely IPL, SHE, and APT [1, pp. 160–162, 180–181].

(S2) The same discussion also says that the output of the multiradial algorithm does not consist of a simple transport of a region by a set-theoretic function, and that the possible regions output by the algorithm cannot automatically be directly compared with codomain-side structures; further techniques are needed [1, p. 161; pp. 133–135].

(S3) In Step (xi-e), after passage to a holomorphic hull and application of the log-volume, the output side is presented as a collection of possible pilot-object log-volumes of the form

$$\mathbb{R}_{\leq -|\log(\Theta)|},$$

linked/related to the input pilot-object log-volume $-|\log(q)|$ [1, p. 183].

(S4) In Step (xi-f), Mochizuki concludes that the above construction of output data possibilities constitutes, possibly up to approximation, a construction of the input pilot-object log-volume, and therefore that the inclusion $-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}$ and the resulting inequality follow formally [1, p. 183].

(S5) In later exposition, Mochizuki describes the same picture as one of noninterference or cohabitation/coexistence of the q-pilot and Θ-pilot in a common container obtained by multiradial representation, holomorphic hull, and log-volume [2, p. 98].

(S6) Scholze and Stix isolate precisely this step as critical and argue that the move from abstract linkage to the required numerical comparison has not been made explicit enough [3, pp. 8–9].

This note studies only the formal implication suggested by (S1)–(S5). It does *not* claim that IUT as a whole is false. It proves only the following: if one keeps the qualitative package and does not add any separate numerical bridge, then the bridge statement is not a logical consequence.

3 Formalization of the qualitative package

We now introduce a minimal abstract framework. The purpose is not to formalize all of IUT, but to isolate the exact logical shape of the disputed implication.

Definition 3.1 (Single-step qualitative package)**.** A *single-step qualitative package* consists of the following data:

(Q1) a nonempty set A of *abstract labels*;

(Q2) a nonempty set C called the *common container*;

(Q3) a projection map $\pi\colon C \to A$;

(Q4) a distinguished element $x_q \in C$ called the *input realization*;

(Q5) a subset $O_\Theta \subseteq C$ called the *output possibilities*;

(Q6) a hull operator $H\colon \mathscr{P}(C) \to \mathscr{P}(C)$, i.e. a map satisfying

$$\begin{aligned} X &\subseteq H(X), \\ X \subseteq Y &\implies H(X) \subseteq H(Y), \\ H(H(X)) &= H(X) \end{aligned}$$

for all $X, Y \subseteq C$;

(Q7) a map $r\colon C \to \mathbb{R}$ called the *log-volume map*;

(Q8) a real number $u \in \mathbb{R}$ such that

$$r\big(H(O_\Theta)\big) = \mathbb{R}_{\leq u} := \{t \in \mathbb{R} \mid t \leq u\}.$$

These data are required to satisfy the *qualitative axioms*

(QA1) **Input real value.** The number

$$L := r(x_q)$$

is a well-defined element of $\mathbb{R}$.

(QA2) **Output half-line.** The image of the hull of the output possibilities under the log-volume map is the closed half-line $\mathbb{R}_{\leq u}$.

(QA3) **Abstract linkage.** Every element of $H(O_\Theta)$ has the same abstract label as the input realization:

$$\pi(c) = \pi(x_q) \qquad \text{for all } c \in H(O_\Theta).$$

(QA4) **Common-container cohabitation.** Both the input realization x_q and all output possibilities lie in the same ambient set C.

Remark 3.2. Definition 3.1 intentionally captures only the qualitative package emphasized in the cited passages. In particular:

- the abstract linkage axiom records that the input and output are related at the level of abstract labels;
- the common-container axiom records the cohabitation/common-container language;
- the output half-line axiom records the shape of the possible output log-volumes;
- no further axiom is imposed that connects the numerical value $L = r(x_q)$ to the set $r(H(O_\Theta))$.

The whole point of the present note is that this final omission is exactly the issue under dispute.

Definition 3.3 (Bridge statement)**.** Given a single-step qualitative package, the associated *bridge statement* is

$$\mathbf{Br}: \qquad L \in \mathbb{R}_{\leq u}.$$

Equivalently, since $\mathbb{R}_{\leq u} = r(H(O_\Theta))$, the bridge statement may be written as

$$L \in r(H(O_\Theta)).$$

Remark 3.4. In the IUT notation motivating this paper, one should think of

$$L = -|\log(q)|, \qquad u = -|\log(\Theta)|.$$

Then the bridge statement is exactly the inclusion used in Step (xi-f).

4 Approximation does not weaken the bridge

Step (xi-f) allows the possibility that the construction of the input log-volume from the output possibilities may hold only "up to some sort of approximation". Because the target set is a closed half-line, this language does not weaken the required bridge. We record the elementary lemma explicitly.

Lemma 4.1 (Closed-half-line approximation lemma)**.** *Let $u, L \in \mathbb{R}$, and let $S := \mathbb{R}_{\leq u}$. Then the following are equivalent:*

(i) $L \in S$.

(ii) *For every $\varepsilon > 0$ there exists $s_\varepsilon \in S$ such that*

$$L - \varepsilon \leq s_\varepsilon.$$

(iii) *For every* $\varepsilon > 0$ *one has*

$$L \leq u + \varepsilon.$$

(iv) $L \leq u$.

Proof. The equivalence (i) $\Leftrightarrow$ (iv) is immediate from the definition of the half-line $S = \mathbb{R}_{\leq u}$.
(i) $\Rightarrow$ (ii). If $L \in S$, then for every $\varepsilon > 0$ we may take $s_\varepsilon := L$. Since $L \in S$ and $L - \varepsilon \leq L$, condition (ii) holds.
(ii) $\Rightarrow$ (iii). Fix $\varepsilon > 0$. By (ii) there is $s_\varepsilon \in S$ with $L - \varepsilon \leq s_\varepsilon$. But $s_\varepsilon \in S = \mathbb{R}_{\leq u}$ implies $s_\varepsilon \leq u$. Hence

$$L - \varepsilon \leq u,$$

so $L \leq u + \varepsilon$.
(iii) $\Rightarrow$ (iv). Assume (iii) and suppose for contradiction that $L > u$. Set

$$\varepsilon := \frac{L - u}{2} > 0.$$

Then (iii) yields

$$L \leq u + \varepsilon = u + \frac{L - u}{2} = \frac{L + u}{2}.$$

Since $L > u$, we have $(L + u)/2 < L$, a contradiction. Thus $L \leq u$.

This proves the equivalence of (i)–(iv). □

Corollary 4.2. *In the setting of Definition 3.1, the following are equivalent:*

(a) *the bridge statement* **Br***;*

(b) *for every* $\varepsilon > 0$ *there exists* $y_\varepsilon \in H(O_\Theta)$ *such that*

$$L - \varepsilon \leq r(y_\varepsilon);$$

(c) $L \leq u$.

Proof. Because $r(H(O_\Theta)) = \mathbb{R}_{\leq u}$, condition (b) is exactly condition (ii) of Lemma 4.1. The equivalence now follows immediately from that lemma. □

Remark 4.3. Corollary 4.2 is the formal reason that the "approximation" parenthesis in Step (xi-f) cannot rescue the disputed implication. In the relevant closed-half-line situation, approximate realization from below is not weaker than the exact bridge; it is equivalent to it.

5 The countermodel and the main negative theorem

We now prove the main result.

Theorem 5.1 (Main negative theorem)**.** *There exists a single-step qualitative package satisfying* (QA1)–(QA4) *for which the bridge statement* **Br** *is false. Consequently, if* $\mathcal{Q}$ *denotes the class of all single-step qualitative packages, then*

$$\mathcal{Q} \nvdash \mathbf{Br}.$$

In words: the bridge is not a logical consequence of the qualitative package alone.

Proof. We construct an explicit package.

Step 1: The ambient sets. Let
$$A := \{*\}$$
be a singleton set. Let
$$C := A \times \mathbb{R} = \{(*, t) \mid t \in \mathbb{R}\}.$$
Define the projection
$$\pi \colon C \to A, \qquad \pi(*, t) := *.$$
Thus all elements of C carry the same abstract label, while their concrete real coordinate may vary arbitrarily.

Step 2: The input realization. Set
$$x_q := (*, 0) \in C.$$
Define the log-volume map
$$r \colon C \to \mathbb{R}, \qquad r(*, t) := t.$$
Then
$$L := r(x_q) = 0.$$
So the input realization has a perfectly well-defined real-valued log-volume.

Step 3: The output possibilities and the hull. Define
$$O_\Theta := \{(*, t) \in C \mid t \leq -1\}.$$
Let the hull operator be the identity operator:
$$H(X) := X \qquad \text{for every } X \subseteq C.$$
This is indeed a hull operator: it is extensive, monotone, and idempotent. Moreover,
$$H(O_\Theta) = O_\Theta.$$
Finally, set
$$u := -1.$$
Then
$$r(H(O_\Theta)) = r(O_\Theta) = \{t \in \mathbb{R} \mid t \leq -1\} = \mathbb{R}_{\leq -1} = \mathbb{R}_{\leq u}.$$
Thus the output possible log-volumes form the required closed half-line.

Step 4: Verification of the qualitative axioms. We check each axiom in Definition 3.1.

(QA1) Since $L = r(x_q) = 0 \in \mathbb{R}$, the input real value axiom holds.

(QA2) We have just computed that
$$r(H(O_\Theta)) = \mathbb{R}_{\leq -1} = \mathbb{R}_{\leq u},$$
so the output half-line axiom holds.

(QA3) Let $c \in H(O_\Theta) = O_\Theta$. Then $c = (*, t)$ for some $t \le -1$. Hence

$$\pi(c) = * = \pi(x_q).$$

So every output possibility has the same abstract label as the input realization. Thus the abstract linkage axiom holds in the strongest possible form: the input and every output possibility are abstractly identical.

(QA4) Both x_q and all elements of O_Θ belong to the same set C by construction. Hence the common-container cohabitation axiom holds.

Step 5: Failure of the bridge. The bridge statement is

$$L \in \mathbb{R}_{\le u}.$$

But here $L = 0$ and $u = -1$, so

$$\mathbb{R}_{\le u} = \mathbb{R}_{\le -1} = (-\infty, -1].$$

Therefore

$$0 \notin (-\infty, -1].$$

Equivalently,

$$L = 0 \notin \mathbb{R}_{\le -1} = r(H(O_\Theta)).$$

So the bridge statement is false.

This constructs a model of the qualitative package in which the bridge fails. Hence $\mathcal{Q} \nvdash \mathbf{Br}$. □

Corollary 5.2 (Failure of approximate bridge in the same model)**.** *In the countermodel of Theorem 5.1, the approximation condition of Corollary 4.2*(b) *also fails.*

Proof. Take $\varepsilon = 1/2$. If there were some $y_{1/2} \in H(O_\Theta) = O_\Theta$ with

$$L - \frac{1}{2} \le r(y_{1/2}),$$

then since $L = 0$ we would have

$$-\frac{1}{2} \le r(y_{1/2}).$$

But every element of O_Θ has the form $(*, t)$ with $t \le -1$, so $r(y_{1/2}) \le -1$, contradiction. Therefore the approximation condition fails.

oindent This is consistent with Corollary 4.2, since in the same model the exact bridge also fails. □

Remark 5.3 (Why the countermodel is structurally relevant)**.** The countermodel is not based on destroying abstract linkage. On the contrary, it maximizes abstract linkage: every object has the same abstract label $*$. Nor does it separate the input and the output into different universes: they lie in the same common container C. The point is exactly that none of these qualitative facts forces the concrete real coordinate of the input to lie in the output half-line. This is precisely the logical gap isolated by the countermodel.

6 What a repaired proof would have to add

The negative theorem identifies the minimal missing ingredient. To deduce the bridge, one must add a statement that genuinely connects the input real value to the output half-line. Several formulations are equivalent in the closed-half-line setting.

Proposition 6.1 (Equivalent repaired bridge axioms)**.** *In the setting of Definition 3.1, the following conditions are equivalent:*

(1) $L \in r(H(O_\Theta))$.

(2) *There exists* $y \in H(O_\Theta)$ *such that* $r(y) = L$.

(3) *For every* $\varepsilon > 0$ *there exists* $y_\varepsilon \in H(O_\Theta)$ *such that*

$$L - \varepsilon \leq r(y_\varepsilon).$$

(4) $L \leq u$.

Proof. The equivalence (1) ⇔ (4) is immediate from the identity $r(H(O_\Theta)) = \mathbb{R}_{\leq u}$. The implication (2) ⇒ (1) is immediate. Conversely, if (1) holds, then by definition of an image set there exists $y \in H(O_\Theta)$ such that $r(y) = L$, so (1) ⇒ (2). The equivalence with (3) is exactly Corollary 4.2. □

Remark 6.2. Proposition 6.1 shows that any successful repair of the Step (xi-e)→Step (xi-f) passage must add a bridge of essentially this strength. In the closed-half-line setting, there is no substantially weaker approximation-based substitute.

7 Conclusion

The note proves a formally precise negative result. If one extracts from the cited IUT passages only the qualitative data that are explicitly highlighted there—abstract linkage, simultaneous qualitatively meaningful construction, passage to a common container, and the fact that the output possible log-volumes form a closed half-line—then the bridge statement

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}$$

does not follow. The proof is the countermodel of Theorem 5.1.

This conclusion is deliberately limited. It does *not* establish that IUT is false. It establishes only that the disputed bridge cannot be recovered from the qualitative package alone. Any complete proof of Corollary 3.12 must therefore use an additional bridge lemma, whether explicit or implicit, that connects the input real-valued log-volume to the output half-line in a way that the qualitative package by itself does not guarantee.

A Correspondence with the motivating notation

For quick reference, the following table records the notational correspondence between the abstract package of this note and the motivating symbols in Step (xi).

Abstract symbol in this note	Motivating IUT notation
$L = r(x_q)$	$-\lvert\log(q)\rvert$
u	$-\lvert\log(\Theta)\rvert$
$r(H(O_\Theta)) = \mathbb{R}_{\leq u}$	Output possible pilot-object log-volumes after hull and log-volume
$\pi(c) = \pi(x_q)$ for $c \in H(O_\Theta)$	Linked/related via abstract prime-strip data
C	Common container obtained by multiradial representation, hull, and log-volume
Br	Inclusion $-\lvert\log(q)\rvert \in \mathbb{R}_{\leq -\lvert\log(\Theta)\rvert}$

References

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[2] Shinichi Mochizuki, *On the Essential Logical Structure of Inter-universal Teichmüller Theory in Terms of Logical AND "∧"/Logical OR "∨" Relations: Report on the Occasion of the Publication of the Four Main Papers on Inter-universal Teichmüller Theory*, March 2024. Available at: `https://www.kurims.kyoto-u.ac.jp/~motizuki/Essential%20Logical%20Structure%20of%20Inter-universal%20Teichmuller%20Theory.pdf`.

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